TITLE: Phan Tuấn Kiệt 2352654 Date: W06

(A)

~~Bayes theorem use the~~

Apply Bayes's probability formula to calculate the probability of correctly predicting a data set when the hypothesis is known.

$P(D|\theta)$ (D: data set; θ: ~~hypothesis~~ ~~parameter~~ parameter value)

the goal is find the θ to maximize value of $P(D|\theta)$.

to find new θ we use bayes theorem

$$P(\theta|D) = \frac{P(D|\theta) \cdot P(\theta)}{P(D)}$$

$P(\theta)$: is prior hypothesis to ~~obtain~~ obtain θ

$P(D)$: evidence

$P(D|\theta)$: likelihood, which mean that with ~~value~~ θ hypothesis θ, how much true able of this hypothesis on D.

$P(\theta|D)$: posterior hypothesis ~~after~~ observe the evidence.

(B)

$X = (x_1, x_2 \dots x_n)$

X is the set of feature, if each feature in X are independence we can calculate $P(X|\theta) = \prod_{i=x}^{X} P(x_i|\theta)$

Similarly we can calculate $P(\theta|X) \propto P(X|\theta) \cdot P(\theta)$

$\propto \prod_{i=x}^{X} P(x_i|\theta) \cdot P(\theta)$

(C)

if one feature have not apperance in train set, when validation set appear that feature can leading to $P(X|\theta) = 0$ this is because ~~the~~ probability of this feature when know $\theta = 0$.

~~So~~ To deal this problem, add each feature ~~in~~ 1 sample and then use laplace smoothing to calculate likelihood, because of the number of this feature different 0 ~~not~~ lead $P(X|\theta) \neq 0$.

(D)

Just write it

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TITLE: Date:

In Generative model, the model Generate the rule that use to ~~look~~ classify sample. While discriminate model make a decision based on the characteristics of the samples.

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